IBM

H. Kolsky
Dept 910
Proj. 022-40,000

SHARE SYMBOLIC CODING FORM

FORM 12-6809-1

PROBLEM

CODER J. Cocke & H Kolsky | DATE May 5, 1958 | PAGE 1 | OF

H	LOCATION		OP		ADDRESS, TAG, DECREMENT	COMMENTS	IDENTIFICATION
1	2 6	7	8 10	11	12 72		73 80
	ST2		LXD		TC, 1		AI HTR OPCODE, 0, -AI+1
	DB		CLA		0, 1	LOC AI IN X1	
			PDX		0, 2	LOC AI+1 IN X2	
			PAX		0, 4	OPCODE IN X4	
			SXD		JB, 2	STORE AI+1	
			TRA		0, 4	TR TO OP	
	AND		TXI		JH1, 1, 3	LOGICAL AND	$-(A_i-3)$
	JH1		SXD		BL1, 1		
			CLA		1, 2	$I_{im} = +1-(-A_{i+1})$	
			PAX		0, 4	$-A_{jm+(2)} \rightarrow X4$	
			CAL		0, 4	C_j or T_j	
			SLW		NG		Temp
			TXI		HM1, 2, -1	$I_{im-1} = (-I_{im})-1$	
	HM1		CLA		1, 2	$+1 + I_{im-1}$	
			PAX		0, 4	$-A_{jm-1+(2)} \rightarrow X4$	
			CAL		NG		
			ANA		0, 4		
			SLW		NG		$\rightarrow b$
			TXI		BL1, 2, -1	$-(A_{i+1+1+m})$	
	BL1		TXH		HM1, 2, 0	$D = -(A_i-3)$ $X = -(A_{i+1}+m)$	
			SLW		0, 2	STORE ANSWER IN OI	
			LXD		JB, 1	AI+1 IN X1	
			TRA		DB		

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H	LOCATION		OP		ADDRESS, TAG, DECREMENT	COMMENTS	IDENTIFICATION
1	2 6	7	8 10	11	12 72		73 80
	OR		TXI		JH2,1,3	LOGICAL OR	
	JH2		SXD		BL2,1		
			CLA		1,2		
			PAX		0,4		
			CAL		0,4		
			SLW		NG		
			TXI		HM2,2,-1		
	HM2		CLA		1,2		
			PAX		0,4		
			CAL		NG		
			ORA		0,4		
			SLW		NG		
			TXI		BL2,2,-1		
	BL2		TXH		HM2,2,0		
			SLW		0,2	STORE ANSWER	
			LXD		JB,1	AI+1 IN X1	
			TRA		JB		
	EXOR		TXI		JH3,1,3	LOGICAL EXCLUSIVE OR	
	JH3		SXD		BL3,1		
			CLA		1,2		
			PAX		0,4		
			CAL		0,4		
			SLW		NG		
			TXI		HM3,2,-1		
	HM3		CAL		1,2		

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H	LOCATION	OP	ADDRESS, TAG, DECREMENT	COMMENTS	IDENTIFICATION
1	2 6	8 10	12 72		73 80
		PAX	0,4		
		CAL	NG		
		ORA	0,4	A+B	
		SLW	LG	Temp.	
		CAL	NG		
		ANA	0,4	A.B	
		COM	0	NOT A.B	
		ANA	LG	(A+B).(NOT A.B)	
		SLW	NG		
		TXI	BL3,2,-1		
	BL3	TXH	HM3,2,0		
		SLW	0,2	STORE ANSWER	
		LXD	JB,1		
		TRA	DB		
	DEL	CAL	-1,2	DELAY (+ACLD)+1	
		PAX	0,1		
		CAL	0,1		
		SLW	+2,2	STORE ANSWER	
		LXD	JB,1		
		TRA	DB		
	ADV	SXD	LG,1	ADVANCE OUTPUTS TO T AND C	Store Loc (AT)
		CAL	-3,2		
		SLW	-2,2	O TO T	
		COM	0		
		SLW	-1,2	NOT O TO C	

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H	LOCATION	OP	ADDRESS, TAG, DECREMENT	COMMENTS	IDENTIFICATION
		CLA	0, 2		
		PDX	0, 2	AI+1 TO X2	X ≤ D
		PXD	0, 2		(Ain) ≤ (-Aadr)
		SUB	LG	TEST IF THRU	
		TNZ	ADV+1		
	OUT	PDX	0, 0	PREPARE PRINT LINE	
		SWT	1	SW 1 DOWN NO PRINT	
		CLA	L1		
		STO	PBIT		
		CLA	L1		
		SWT	2	SW2 DOWN NO TAPE	
		CLA	L0		
		STO	TBIT		
		TZE	JC		
		CLA	PBIT		
		TZE	DEC	SKIP ALL I/O	
	JC	EXD	0, 0	SET UP PRINT LINE	
		SLW	NG		
		LXD	L6, 2		
		LXD	L0, 5		
				PE CLA PTBL, 1	
	SE	CLA	0	SUB TC2	
		TZE	BE	TMI PE1	
		LDQ	ONE	SUB LX2	
	SET	CAL	NG	PE1 ADD TC2	
				STA SE	

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H	LOCATION	OP	ADDRESS, TAG, DECREMENT	COMMENTS	IDENTIFICATION
		LGL	6		
		SLW	NG		
		TIX	HE, 2, 1		
		SLW	PBLK, 4	STORE PRINT WORD	
		TXI	JE, 4, 1		
	JE	TXH	HK, 4, 19	TR OUT TO PRINT	
		LXD	L6, 2		
	HE	CLA	PTBL+1, 1		
		TZE	BET	TEST IF END OF PRINT TABLE	
		TXI	PE, 1, 1		
	BE	LDQ	ZERO		
		TRA	SET		
	BET	LDQ	BLNKS		
		TRA	SET		
	HK	CLA	MARK		
		ADD	L1		
		STO	NG		
		PXD	0, 0		
		LDQ	NG		
		DVH	L10		
		STO	MARK		
		PDX	0, 1		
		CAL	PBLK+19		
		LRS	6		
		LDQ	MKBLK, 1		

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H	LOCATION	OP	ADDRESS, TAG, DECREMENT	COMMENTS	IDENTIFICATION
		LGL	6		
		SLW	PBLK+19		
		CLA	PHDB		
		TNZ	PLINE		
		TSX	WOT,4	PRINT TITLE	
			HD1,0,HD1+9		
		TSX	WOT,4	SPACE	
			BLNKS,0,BLNKS		
	PLINE	TSX	WOT,4	PRINT LINE	
			PBLK,0,PBLK+19		
	(DEC follows here)				
	START	STZ	MARK	INITIALIZE START OF PROB	
		CLA	TC		
	ST1	PDX	0,1	SET COMPLEMENTS	
		CLA	0,1		
		TAX	0,4	TEST OPCODE FOR ADV	
		PXA	0,4		
		SUB	TC1		
		TZE	ST2	TR WHEN THRU	
		CAL	-2,1		
		COM	0		
		SLW	-1,1		
		CLA	0,1		
		TRA	ST1	AI+1 TO X1	

(ST2 follows here)

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H	LOCATION		OP		ADDRESS, TAG, DECREMENT	COMMENTS	IDENTIFICATION
1	2 6	7	8 10	11	12 72		73 80
	RCD1		RCD		READ CARD READER		
			CPY		IB		
			CPY		IB+1	LXD L35, 2	
			LXD		L0, 1	SXD RCD3, 2	
			LDQ		IB		
	RCD2		PXD		0, 0		
			LGL		1		
			TZE		RCD4		
			CAL		Z1		
	RCD5		SLW		B0, 1	SET INPUT SWITCHES	
			TXI		RCD3, 1, 1		
	RCD3		TXL		RCD2, 1, 0	PXD 0, 1	
			LDQ		IB+1	SUB L72	
			LXD		L71, 2	TZE ST2 TR TO START	
			SXD		RCD3, 2		
			TRA		RCD2		
	RCD4		CAL		L0		
			TRA		RCD5		
	DEC		CLA		TNI	SIGNAL FOR NEW INPUT (temporary)	
			TNZ		ST2		
			CLA		L1	SW 3 DOWN TO READ CARD	
			SWT		3		
			CLA		L0		
			STO		TNI		
			SWT		3		
			TRA		ST2	TR TO START	

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H	LOCATION	OP	ADDRESS, TAG, DECREMENT	COMMENTS	IDENTIFICATION
	L1	HTR	0,0,1		
	L0	HTR	0		
	TBIT	BSS	1		
	PBIT	BSS	1		
	LG	BSS	1		
	NG	BSS	1		
	JB	BSS	1		
	ONE	BCD	11		
	ZERO	BCD	10		
	L6	HTR	0,0,6		
	LX2	HTR	2		
		BCD	19		
		BCD	18		
		BCD	17		
		BCD	16		
		BCD	15		
		BCD	14		
		BCD	13		
		BCD	12		
		BCD	11		
	MKBLK	BCD	10		
	PBLK	BSS	20		
	TC	HTR	0,0,-A1		
	HD1	BCD	0PROJ 7000 · LOGIC SIMULATOR COCKE-NOLSKY MAY 58		
	Z1	OCT	-377777777		
	TC1	HTR	ADV		

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1	2 6	7	8 10	11	12 72		73 80
	IB		BSS		2		
	BO		BES		73		
	L35		HTR		0, 0, 35		
	L72		HTR		0, 0, 72		
					PROJ 7000 LOGIC SIMULATOR COCKE & KOLSKY MAY 58		
			ORG		100		
	TNI		BSS		1		
	TC2		HTR		BO		
	PTBL		BES		120		